

Failure Case F-007: Slide Mechanics Expose LLM Limits

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Trigger	Slide-mechanic levels consistently harder for LLM to design solvably than push levels.
What happened	Push solvability: 90% (9/10 levels) Slide solvability: 71% (5/7 levels) The Level Designer does not fully account for slide physics when placing boxes and targets. In slide mode, a pushed box slides until hitting a wall or another box. Boxes that slide past targets (no wall stopper behind the target) create unsolvable layouts. Baseline comparison confirms this is an LLM spatial reasoning limitation: Single-LLM baseline slide solvability: 30% (3/10) Multi-agent pipeline slide solvability: 71% (5/7) The debug loop more than doubles slide solvability, but cannot fully compensate.
Root cause	LLMs cannot reliably reason about sliding physics. Push mechanics are straightforward (box moves exactly 1 tile), but slide mechanics require reasoning about distances, wall positions, and chain reactions. The Level Designer's spatial reasoning breaks down when boxes move more than one tile.
Mitigation applied	Added slide-specific design guidance to Level Designer prompt: - Place targets adjacent to walls (so sliding boxes stop on them) - Use interior walls as stoppers - Avoid wide open spaces where boxes slide uncontrollably This improved slide solvability from ~30% (baseline) to 71% (pipeline), but the gap with push (90%) remains.
Governance lesson	When an LLM agent has a systematic weakness, the system should compensate with stronger verification and more repair cycles -- not rely on prompt engineering alone. The BFS solver catches every unsolvable slide level; the debug loop fixes most of them. The remaining gap suggests a need for solver-guided generation, not just solver-verified generation.